

Nagorik Sheba

Agentic Municipal Complaint Management
with Citizen Participation

Project Repository:

MozazaZaman/cseku_26_wpl_nagorik_sheba

Status:  Working on mobile app currently

Tests: 150+ Unit Tests

Coverage: Database • Agents • Auth

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1 Project Overview

Executive Summary

Nagorik Sheba is an *Agentic Web and Mobile Platform* designed for transparent municipal complaint management in Bangladesh. Citizens can report issues (potholes, power outages, water leaks) and track resolution in real-time through an AI-powered pipeline.

1.1 Key Differentiators

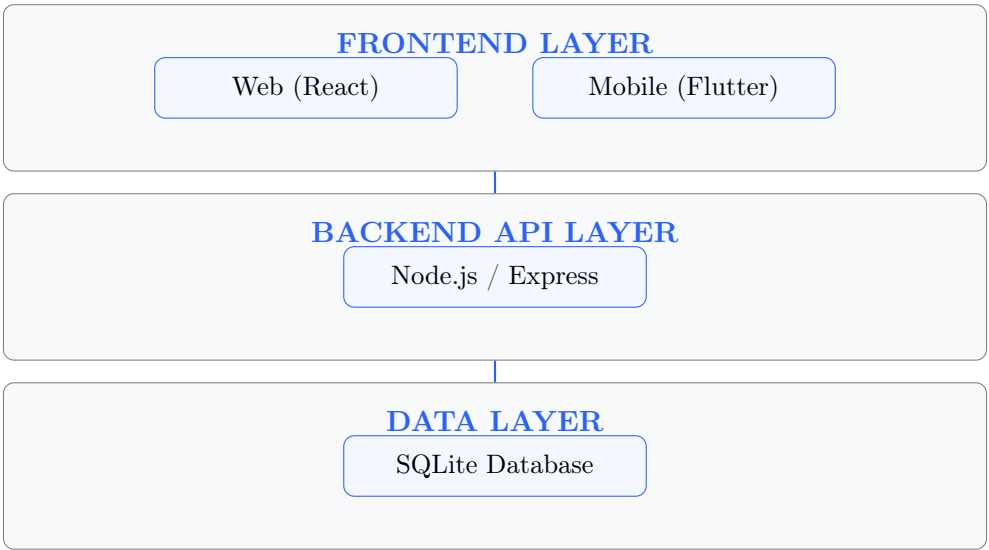
- ✓ Transparent, AI-driven complaint processing
- ✓ Citizen participation through voting system
- ✓ Multilingual support (English, Bangla, Roman transliteration)
- ✓ Geographic routing to correct authorities
- ✓ Real-time tracking and notifications

1.2 Technology Stack

Layer	Technology	Purpose
Backend	Node.js + Express	REST API server
Database	SQLite	Lightweight data storage
Mobile	Flutter + Dart	Cross-platform app
Web	React + JavaScript	Web dashboard
Testing	Vitest	Unit test framework
Auth	JWT + bcryptjs	Secure authentication

2 System Architecture

2.1 Three-Tier Stack



3 The 5-Agent Complaint Pipeline

Architecture Insight

Every complaint passes through 5 intelligent agents before reaching the appropriate authority. This ensures quality, prevents spam, and routes issues correctly.

1. Agent 1: Photo Verifier

- Validates complaint content (min 10 characters)
- Verifies image authenticity (1KB–8MB)
- Detects spam patterns
- *Tests: 9 unit tests*

2. Agent 2: Classifier

- Categorizes into 6 types: Road, Electricity, Water, Gas, Sanitation, Other
- **Multilingual:** English, Bangla, Roman transliteration
- *Tests: 12 unit tests*

3. Agent 3: Duplicate Finder

- Detects similar complaints within 250m GPS radius
- Prevents spam and consolidates votes

4. Agent 4: Priority Ranker

- Scores complaints: $(votes \times 4) + (severity \times 12) + (age \times 0.3)$
- Capped at 100 points
- Detects emergency keywords

5. Agent 5: Destination Router

- Routes to correct authority (City Corp, Pouroshova, Union Parishad)
- Uses GPS coordinates for geographic bound matching

4 Database Schema

4.1 Core Tables

Table	Key Fields
Users	user_id, full_name, email, phone, password_hash, role, face_verified
Authorities	authority_id, name, type, geographic bounds, division, district
Complaints	complaint_id, user_id, authority_id, title, description, category, status
Votes	vote_id, user_id, complaint_id (unique combination)
Agent Logs	agent_name, decision, input_summary, output_summary
Notifications	complaint_id, title, message, is_read

5 Security Features

Security Highlights

The platform implements defense-in-depth security across all layers.

Password Security

- bcryptjs hashing (10 rounds)
- Salted & irreversible
- Unique hash per password

Data Protection

- Parameterized SQL queries
- Sensitive data masking
- Server-only face photos

Authentication

- JWT token-based auth
- Role-based access control
- Session management

Authorization

- Staff jurisdiction enforcement
- Department-level access
- Geographic boundaries

6 Test Coverage

Component	Tests	Status
Database Operations	50+	✓ Passing
Agent Pipeline	35+	✓ Passing
Authentication	40+	✓ Passing
Security	15+	✓ Passing
Multilingual	12+	✓ Passing
Total	150+	Production Ready

6.1 Running Tests

```

1 cd server
2 npm install
3 npm test                # Run once
4 npm test -- --watch    # Watch mode
5 npm run test:coverage  # With coverage report

```

Listing 1: Test Commands

7 Unique Features

Multilingual Support

Handles English ("street light not working"), Bangla (" "), and Roman transliteration ("rasta vanga") seamlessly.

Geographic Routing

GPS coordinates automatically route complaints to the correct municipal authority based on geographic bounds.

📋 Priority Algorithm

Dynamic scoring: $\text{votes} \times 4 + \text{severity} \times 12 + \text{age} \times 0.3$, capped at 100.

✅ Citizen Voting

Democratic priority escalation with deduplication to prevent gaming.

👁️ Transparent Processing

Every agent decision is logged for full auditability.

8 Conclusion

Project Impact

Nagorik Sheba demonstrates a complete full-stack civic-tech solution combining **AI agent orchestration**, **multilingual NLP**, **geographic routing**, and **comprehensive testing** to create a transparent, efficient municipal complaint management system for Bangladesh.

Repository: https://github.com/MozazaZaman/cseku_26_wpl_nagorik_sheba